PHYSICAL CHEMISTRY CUMLATIVE EXAM

September 21, 1991

Thermodynamics of perfect gases:

1) a) What is an isotherm?

b) What is an adiabat?

c) Is the work done during the isothermal reversible expansion of a perfect gas more or less than the work for the adiabatic reversible expansion of the same gas between the same initial and final volumes? (The comparison is for expansions starting from the same initial state.)

(Your answers must be explained. Include a graph with your definitions.)

2) A sample of 3.0 mol of a perfect gas at 200 K and 2.0 atm is compressed reversibly and adiabatically until the temperature reaches 250 K. Given that its molar constant-volume heat capacity is 27.5 J K^{-1} mol^{-1}, calculate the heat transferred to the system q, the work done w, the change in internal energy ΔU, in enthalpy ΔH, and the final pressure and volume.

Help: For reversible adiabatic expansion: $C_V dT = -pdV$; C_V can be considered constant in the given T range.

R = 8.206E-2 L atm /K /mol = 8.314 J/K/mol